

Store Sales Analysis — SQL Analysis

1. Project Objective

The objective of this project is to analyze Store's sales data using SQL and identify meaningful business insights related to sales performance, customer demographics, product categories, sales channels, and geographical performance.

The analysis focuses on answering key business questions such as:

- What is the total sales revenue generated?
- What is the average order value?
- Which product categories generate the highest sales?
- Which sales channels contribute the most revenue?
- How does sales performance differ between male and female customers?
- Which age groups contribute the most to sales?
- Which states generate the highest sales?
- How does sales performance change month by month?
- Which states and categories are the top performers?

The analysis demonstrates the use of SQL for data aggregation, filtering, grouping, ranking, and deriving business-oriented insights from a retail dataset.

2. Dataset Overview

The dataset contains retail order information from a Store. Each record contains information about an order, customer, product, sales amount, quantity, sales channel, and shipping location.

Dataset contains information about:

Column	Description
Order ID	Unique identifier of the order
Customer ID	Identifier of the customer
Gender	Gender of the customer
Age	Age of the customer
Age Group	Customer age classification
Date	Order date
Status	Order status
Channel	Sales channel through which the order was placed

Column	Description
SKU	Product stock keeping unit
Category	Product category
Size	Product size
Qty	Quantity purchased
Currency	Currency of the transaction
Amount	Sales amount
Ship City	Customer shipping city
Ship State	Customer shipping state
Ship Postal Code	Customer postal code
Ship Country	Customer country
B2B	Indicates whether the order is a business transaction

Dataset Size

The dataset contains approximately **31,000+ records** covering retail transactions.

The cleaned dataset was imported into MySQL as the table:

```
cleaned_dataset
```

The `Amount` column was used as the primary measure for sales/revenue analysis.

3. Data Preparation

Before performing SQL analysis, the original dataset was cleaned and prepared for analysis.

Data cleaning activities performed:

1. Gender standardization

Inconsistent gender values such as `m` and `w` were standardized to:

- Men
- Women

This ensures consistent grouping during analysis.

2. Quantity conversion

Text-based quantity values such as:

- one
- two

were converted into numerical values so that total quantity could be calculated correctly.

3. Age Group creation

A new `Age Group` column was created to categorize customers based on age:

- **Teenager:** Age < 30
- **Adult:** Age 30–60
- **Senior:** Age > 60

This makes demographic analysis easier.

4. Month creation

A month field was derived from the order date to analyze monthly sales trends.

5. Column name standardization

Column names containing spaces or special characters were renamed where necessary to make them easier to work with in MySQL.

For example:

```
Order ID → Order_ID
```

6. Data import

After cleaning, the dataset was imported into MySQL as:

```
cleaned_dataset
```

SQL queries were then used to perform the analysis.

4. SQL Queries & Results

Query 1 — Overall Sales KPIs

Business Question

What are the total sales, total orders, total quantity sold, and average order value?

SQL Query

```
SELECT
    SUM(Amount) AS Total_Sales,
    COUNT(DISTINCT Order_ID) AS Total_Orders,
    SUM(Qty) AS Total_Quantity,
    ROUND(SUM(Amount) / COUNT(DISTINCT Order_ID), 2) AS Average_Order_Value
FROM cleaned_dataset;
```

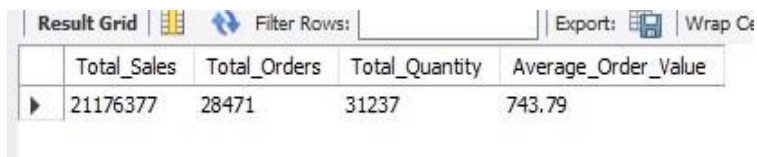
Result

Total Sales: ₹ 21,176,377

Total Orders: 28471

Total Quantity: 31237

Average Order Value: ₹ 743.79



	Total_Sales	Total_Orders	Total_Quantity	Average_Order_Value
▶	21176377	28471	31237	743.79

Query 2 — Sales by Category

Business Question

Which product categories generate the highest sales?

SQL Query

```
SELECT
    Category,
    SUM(Amount) AS Total_Sales
FROM cleaned_dataset
GROUP BY Category
ORDER BY Total_Sales DESC;
```

Result

Top-performing category: Set

Sales: ₹ 10507546

Result Grid			Filter Rows:
	Category	Total_Sales	
▶	Set	10507546	
	kurta	4959377	
	Western Dress	3148836	
	Top	1186199	
	Saree	1010471	
	Ethnic Dress	195256	
	Blouse	140888	
	Bottom	27804	

Query 3 — Sales by Channel

Business Question

Which sales channels generate the highest revenue?

SQL Query

```
SELECT
    Channel,
    SUM(Amount) AS Total_Sales
FROM cleaned_dataset
GROUP BY Channel
ORDER BY Total_Sales DESC;
```

Result

Top-performing channel: **Amazon**

Sales: ₹ **7519933**

Result Grid			Filter Rows:
	Channel	Total_Sales	
▶	Amazon	7519933	
	Myntra	4941540	
	Flipkart	4573301	
	Ajio	1331427	
	Nalli	1015329	
	Meesho	927606	
	Others	867241	

Query 4 — Sales by Gender

Business Question

How do sales differ between male and female customers?

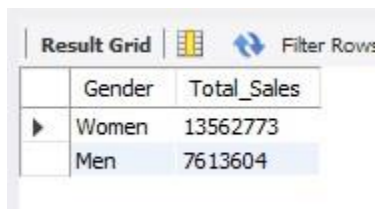
SQL Query

```
SELECT
    Gender,
    SUM(Amount) AS Total_Sales
FROM cleaned_dataset
GROUP BY Gender
ORDER BY Total_Sales DESC;
```

Result

Highest sales contribution: **Women**

Sales: ₹ **13562773**



The screenshot shows a 'Result Grid' window with a 'Filter Rows' button. It contains a table with two columns: 'Gender' and 'Total_Sales'. The table has two rows: 'Women' with a total sales of 13562773, and 'Men' with a total sales of 7613604. The 'Women' row is highlighted with a blue background.

Gender	Total_Sales
Women	13562773
Men	7613604

Query 5 — Sales by Age Group

Business Question

Which customer age group contributes the most to sales?

SQL Query

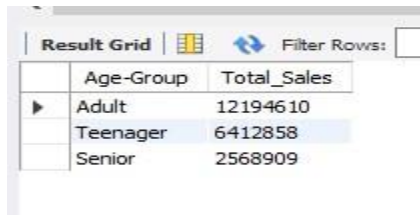
```
SELECT
    Age_Group,
    SUM(Amount) AS Total_Sales
FROM cleaned_dataset
```

```
GROUP BY Age_Group
ORDER BY Total_Sales DESC;
```

Result

Highest-performing age group: Adult

Sales: ₹ 12194610



	Age-Group	Total_Sales
▶	Adult	12194610
	Teenager	6412858
	Senior	2568909

Query 6 — Monthly Sales

Business Question

How does the store's sales performance change from month to month?

SQL Query

```
SELECT
    Month_Number,
    Month,
    SUM(Amount) AS Total_Sales
FROM cleaned_dataset
GROUP BY Month_Number, Month
ORDER BY Month_Number;
```

Result

Highest-sales month: March

Sales: ₹ 1928066

Lowest-sales month: November

Sales: ₹ 1615356

Result Grid			Filter Rows:
	Month	Total_Sales	
▶	April	1829263	
	August	1808505	
	December	1622033	
	February	1875932	
	January	1820601	
	July	1772300	
	June	1750966	
	March	1928066	
	May	1797822	
	November	1615356	
	October	1666662	
	September	1688871	

Query 7 — Top 5 States by Sales

Business Question

Which states generate the highest sales revenue?

SQL Query

```
SELECT
    ship_state,
    SUM(Amount) AS Total_Sales
FROM cleaned_dataset
GROUP BY ship_state
ORDER BY Total_Sales DESC
LIMIT 5;
```

Result

Top state: MAHARASTRA

Sales: ₹ 2990221

<		
Result Grid		
Filter Rows:		
	State	Total_Sales
▶	MAHARASHTRA	2990221
	KARNATAKA	2646358
	UTTAR PRADESH	2104659
	TELANGANA	1712439
	TAMIL NADU	1678877

Query 8 — Sales Contribution by Category

Business Question

What percentage of total sales is contributed by each product category?

SQL Query

```
SELECT
    Category,
    SUM(Amount) AS Total_Sales,
    ROUND(
        SUM(Amount) * 100.0 /
        (SELECT SUM(Amount) FROM cleaned_dataset),
        2
    ) AS Sales_Percentage
FROM cleaned_dataset
GROUP BY Category
ORDER BY Total_Sales DESC;
```

Result Grid			
Filter Rows:			
Export:			
	Category	Total_Sales	Sales_Percentage
▶	Set	10507546	49.62
	kurta	4959377	23.42
	Western Dress	3148836	14.87
	Top	1186199	5.60
	Saree	1010471	4.77
	Ethnic Dress	195256	0.92
	Blouse	140888	0.67
	Bottom	27804	0.13

Query 9 — State Ranking Using RANK()

Business Question

How can states be ranked according to their sales performance?

SQL Query

```
SELECT
    ship_state,
    Total_Sales,
    RANK() OVER (ORDER BY Total_Sales DESC) AS Sales_Rank
FROM (
    SELECT
        ship_state,
        SUM(Amount) AS Total_Sales
    FROM cleaned_dataset
    GROUP BY ship_state
) AS State_Sales;
```

ship-state	Total_Sales	Sales_Rank
MAHARASHTRA	2990221	1
KARNATAKA	2646358	2
UTTAR PRADESH	2104659	3
TELANGANA	1712439	4
TAMIL NADU	1678877	5
DELHI	1266328	6
KERALA	1008940	7
WEST BENGAL	922444	8
ANDHRA PRADESH	918499	9
HARYANA	813320	10
GUJARAT	715563	11
MADHYA PRADESH	564026	12
RAJASTHAN	547360	13
BIHAR	446831	14
ODISHA	414840	15
PUNJAB	368940	16
UTTARAKHAND	327179	17
ASSAM	326423	18
JHARKHAND	255054	19

5. Key Insights

Based on the SQL analysis, the following key business insights were identified:

1. Overall Sales Performance

The store generated total sales of ₹ **21176377** across **28471 distinct orders**, with an average order value of approximately ₹ **743.79**.

2. Category Performance

Set was the highest-performing product category, generating approximately ₹ **10507546** in sales.

3. Channel Performance

Amazon generated the highest sales among the available sales channels, contributing approximately ₹ **7519933**.

4. Customer Demographics

The **Women** customer segment generated higher sales than the other gender segment.

Among the age groups, **Adult** contributed the highest sales.

5. Geographical Performance

MAHARASTRA was the highest-performing state based on total sales.

6. Monthly Sales Trend

Sales were highest in **March** and lowest in **November**.

7. Business Implications

The analysis suggests that the business can potentially focus on:

- Strengthening high-performing product categories.
- Prioritizing the most effective sales channels.
- Targeting major customer age groups with relevant campaigns.
- Focusing marketing and promotional efforts on high-performing states.
- Investigating the reasons behind unusually high or low monthly sales.

6. Tools Used

Microsoft Excel

Used for:

- Initial data cleaning
- Data transformation
- Creating Age Groups
- Creating month-related fields
- PivotTables and PivotCharts
- Interactive dashboard creation using slicers
- Exploratory analysis

MySQL

Used for:

- Loading the cleaned dataset
- Data aggregation
- Filtering and sorting
- GROUP BY analysis
- Aggregate functions such as SUM () and COUNT ()
- COUNT (DISTINCT)
- Subqueries
- Percentage calculations
- RANK () window function
- Business-oriented sales analysis

Microsoft Word

Used to document:

- SQL queries
- Query outputs
- Analysis methodology
- Key business insights

Conclusion

The Store Sales Analysis project demonstrates the complete workflow of a basic retail data analysis project, starting from data cleaning and preparation to SQL-based analysis and dashboard visualization.

Excel was used for data preparation and exploratory analysis, while MySQL was used to perform structured business analysis and extract meaningful insights from the dataset. The analysis examined sales performance across products, channels, customer demographics, states, and months.